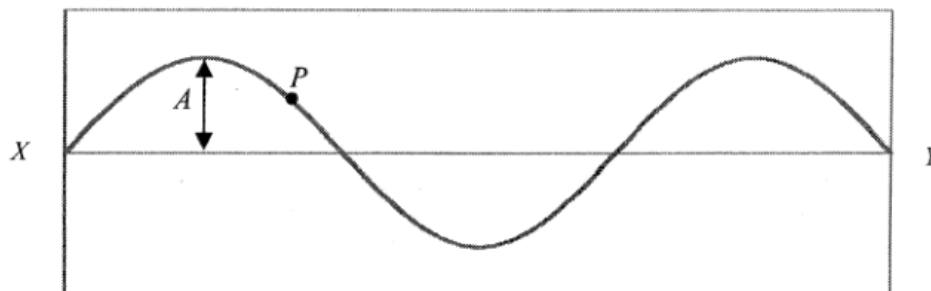


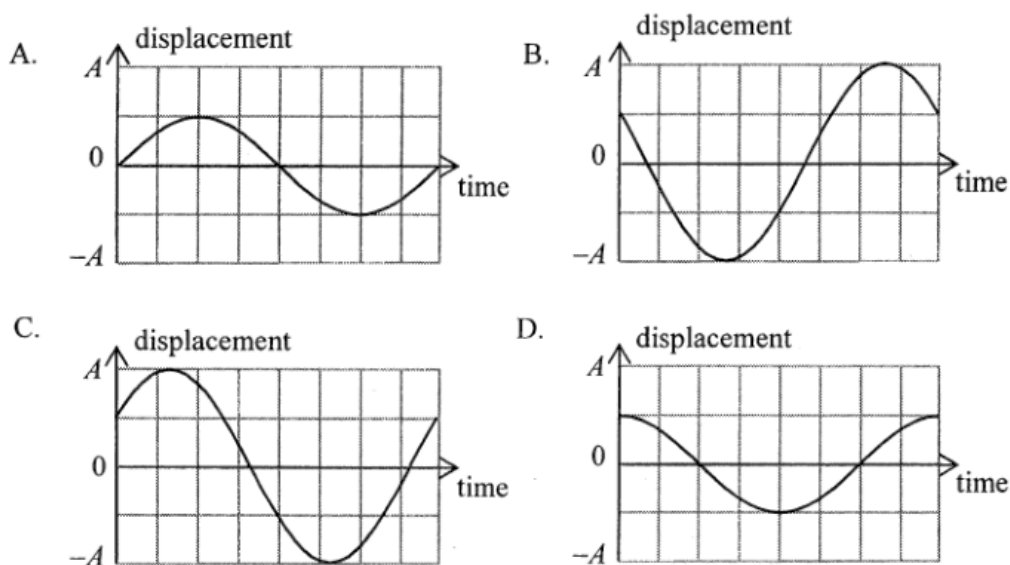
ch15 Interference and Stationary Wave

2012 Q18

18. A stationary wave is formed on a string fixed at both ends X and Y . The following is a snapshot of the string at time $t = 0$. The amplitude of vibration at an antinode is A .

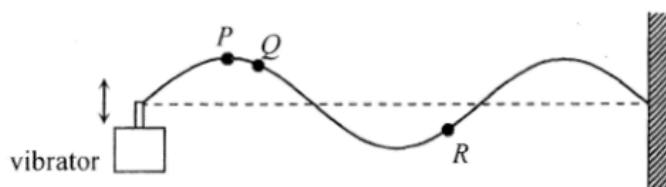


Which of the following shows the displacement-time graph of point P on the string for one period? (Upward displacement is taken as positive.)



2013 Q18

18.

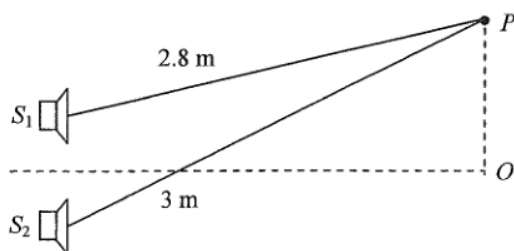


A vibrator generates a stationary wave on a string which is fixed at one end. The figure shows the appearance of the string at a certain instant. Which of the following descriptions about the motion of particles P , Q and R must be correct?

- (1) P and Q are momentarily at rest at this instant.
- (2) Q and R take the same time to reach their respective equilibrium positions.
- (3) P and R are always in antiphase.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

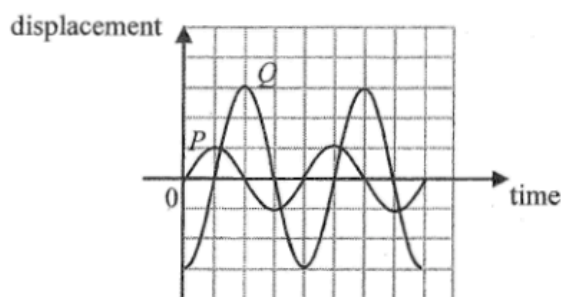
18.



S_1 and S_2 are two loudspeakers connected to a signal generator but the sound waves produced by them are in anti-phase. Point O is equidistant from the loudspeakers while point P is at the distances shown in the figure from the loudspeakers. What type of interference occurs at O and P if the wavelength of the sound waves is 10 cm?

	O	P
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

13.

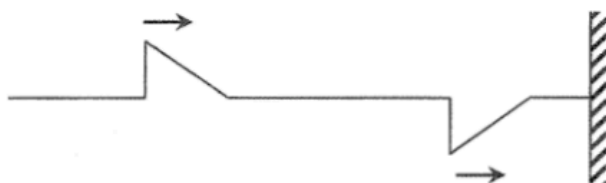


Two waves P and Q travel in the same direction and meet at a point. The above graph shows the variation of the displacement of each wave with time at that point. Which of the following statements is/are correct?

- (1) P and Q have the same frequency.
- (2) The oscillation due to P is in anti-phase with that due to Q .
- (3) The amplitude of the resultant wave at that point is four times the amplitude of P .

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

14.



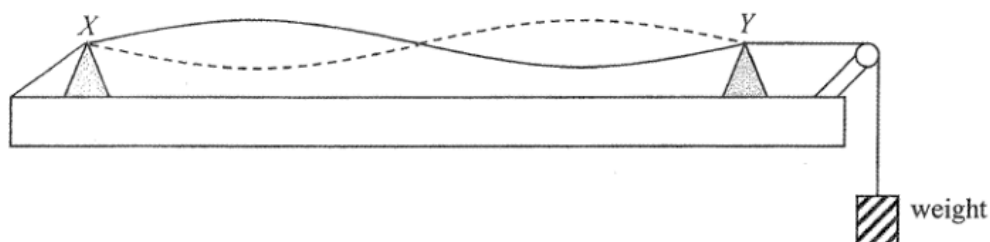
Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above. Which of the following can be the resultant waveform at different instants later?



- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

2015 Q18

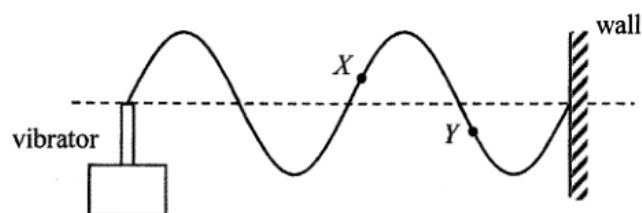
18.



A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown. If the tension in the string is increased by adding weight gradually while the frequency is kept at f , which of the following is a possible mode of vibration at a steady state?

- A.
- B.
- C.
- D.

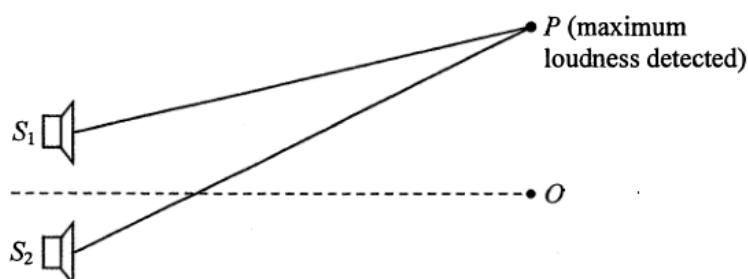
18. A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown.



Which statement is correct when the frequency of the vibrator doubles ?

- A. The wavelength will double.
- B. The wave speed will double.
- C. The amplitude will be halved.
- D. Particles X and Y will become vibrating in phase.

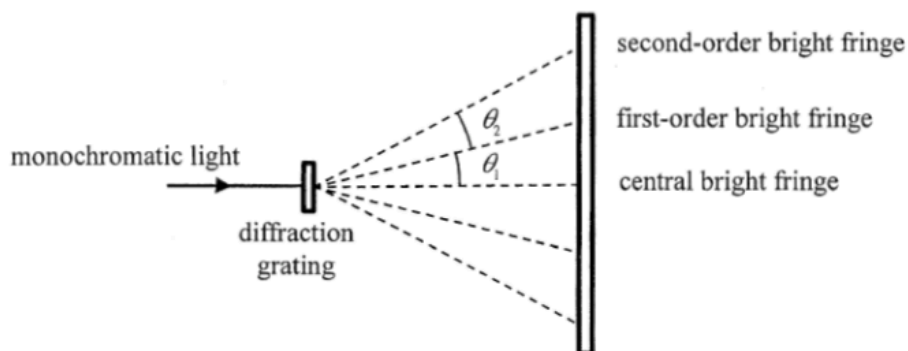
21.



Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ . Which statement is **INCORRECT** ?

- A. Both PS_1 and PS_2 must be integral multiples of wavelength λ .
- B. The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.
- C. At least one point of minimum loudness can be detected between O and P .
- D. Minimum loudness will be detected at P if the sound waves from S_1 and S_2 are in antiphase.

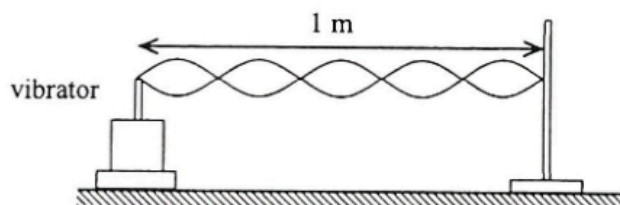
- *20. The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.



Which of the following is/are correct ?

- (1) $\theta_1 = \theta_2$
 - (2) The maximum order of bright fringe is 4 if $\theta_1 = 20^\circ$.
 - (3) θ_1 will decrease if the experiment is performed in water but not in air.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

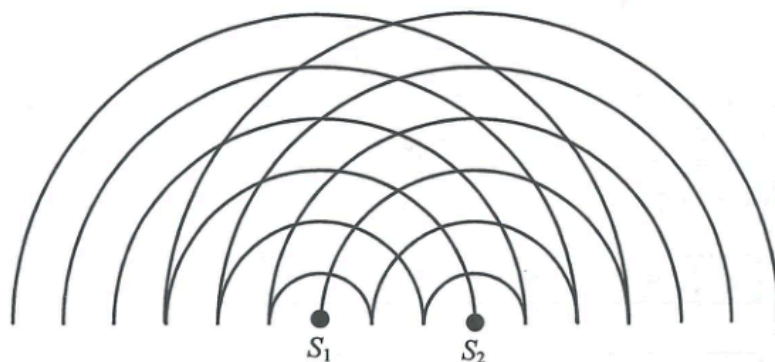
18. The figure shows a string with one end fixed and the other end tied to a vibrator. A stationary wave is formed as shown at a certain frequency.



If the speed of the wave along the string is 7 m s^{-1} , what is the frequency of the wave ?

- A. 2.8 Hz
- B. 7 Hz
- C. 17.5 Hz
- D. 35 Hz

16.

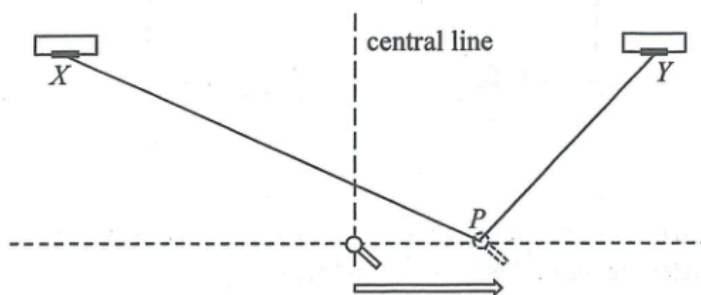


The figure shows the circular water waves generated by two dippers S_1 and S_2 vibrating in phase. The lines represent wave crests. What is the number of nodal lines (i.e. minimum amplitude) formed?

- A. 3
- B. 4
- C. 6
- D. 7

2019 Q21

21.

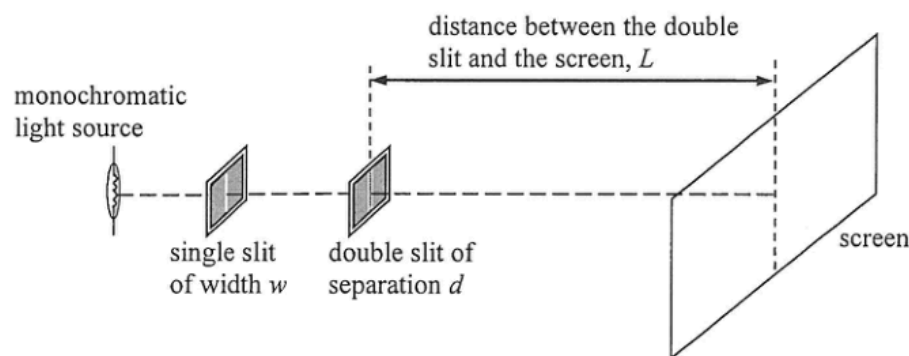


Two loudspeakers X and Y emit sound waves of frequency 500 Hz. A microphone is moved steadily along a line perpendicular to the central line as shown. It detects sound waves of maximum amplitude at the central line and the next maximum amplitude is detected at point P . Find $PX - PY$.

Given: speed of sound in air = 340 m s^{-1}

- A. 0.17 m
- B. 0.34 m
- C. 0.51 m
- D. 0.68 m

15. The figure shows a typical set-up of Young's double slit experiment.

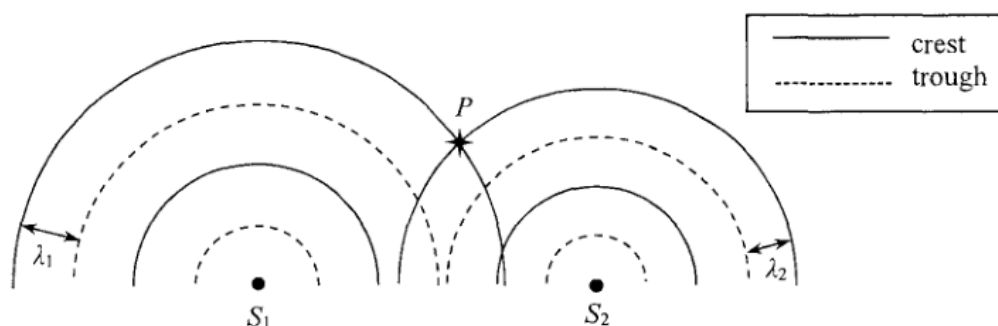


Which combination below is the best setting for displaying an observable fringe pattern on the screen ?

	w	d	L
A.	0.1 mm	1 mm	10 m
B.	0.1 mm	1 mm	1 m
C.	1 mm	0.1 mm	1 m
D.	1 mm	0.1 mm	0.1 m

2021 Q14

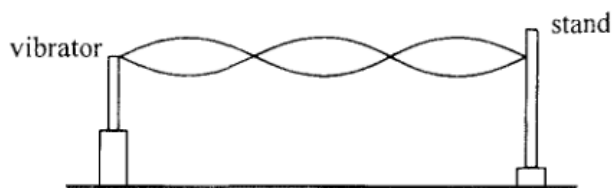
14. In a ripple tank, circular water waves of wavelengths λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. The figure represents the two circular waves propagating on the water surface at a certain moment.



Which of the following statements is correct ?

- A. The particle at P is always at crest position.
- B. At P , the two waves always reinforce to give a larger amplitude.
- C. The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.
- D. The principle of superposition can be applied at P but the two waves do not always reinforce at that location.

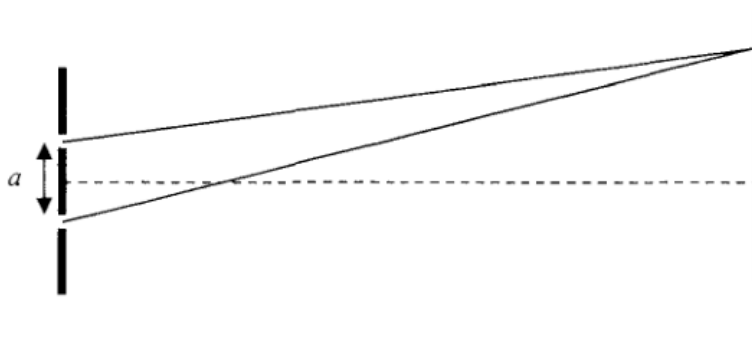
15. In the set-up below, different stationary wave patterns are formed on an elastic string by adjusting the frequency f of the vibrator.



Which statements are correct when f increases?

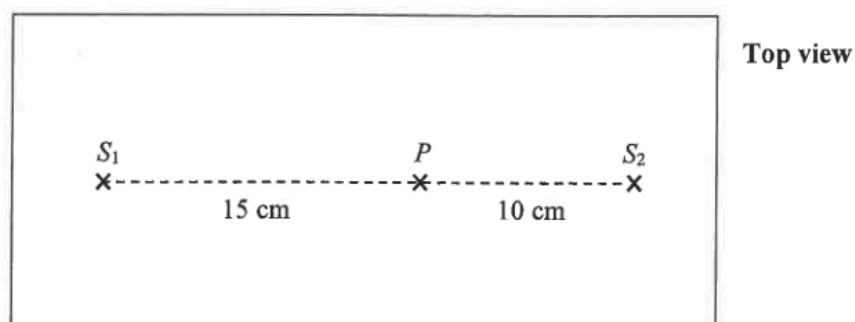
- (1) The number of antinodes increases.
 - (2) The speed of the waves on the string increases.
 - (3) The frequency of the waves produced in air by the string increases.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

16. In Young's double slit experiment employing monochromatic light, how will the interference pattern change if the separation a of the double slit is reduced?



- (1) The pattern will become brighter.
 - (2) The number of fringes that can be observed will increase.
 - (3) The fringe separation of the pattern will become larger.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

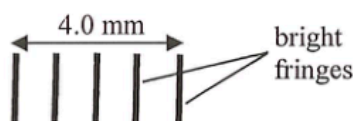
19. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase with frequency f and generate two sets of waves towards each other at a speed of 20 cm s^{-1} . Destructive interference occurs at point P on the straight line S_1S_2 .



Which of the following is a possible value of f ?

- A. 24 Hz
- B. 20 Hz
- C. 18 Hz
- D. 16 Hz

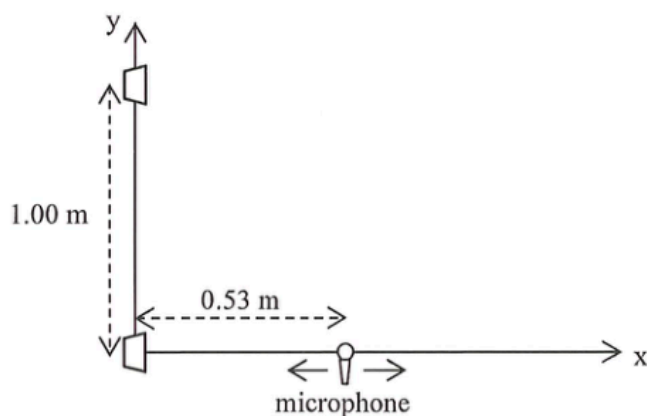
*17.



In Young's double-slit experiment, monochromatic light of wavelength λ is used and the slit separation is a . On a screen placed at a distance of 1.8 m from the double slits, the separation between five consecutive bright fringes is 4.0 mm . Which relationship below is correct?

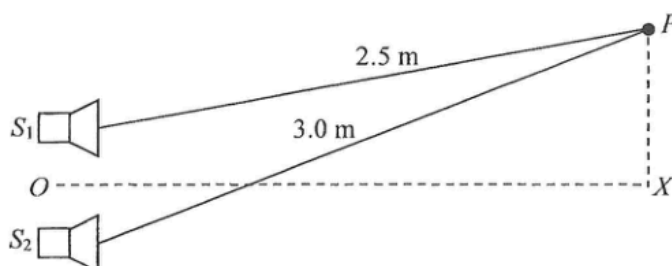
- A. $a = 1800 \lambda$
- B. $a = 2250 \lambda$
- C. $a = 7200 \lambda$
- D. $a = 11250 \lambda$

19. Two loudspeakers separated by 1.00 m along the y-axis emit coherent sound waves which are in phase. When a microphone is moved along the x-axis, only one maximum is detected at a distance of 0.53 m as shown. Find the wavelength of the sound waves.



- A. 0.60 m
 B. 0.75 m
 C. 1.00 m
 D. 1.20 m

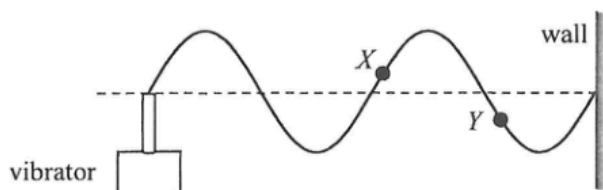
17. Two identical loudspeakers S_1 and S_2 connected to a signal generator produce sounds of wavelength 0.10 m which are in **anti-phase**.



OX is the perpendicular bisector of the line joining S_1 and S_2 . What type of interference occurs at X and P respectively?

- | | X | P |
|----|--------------|--------------|
| A. | destructive | constructive |
| B. | constructive | constructive |
| C. | destructive | destructive |
| D. | constructive | destructive |

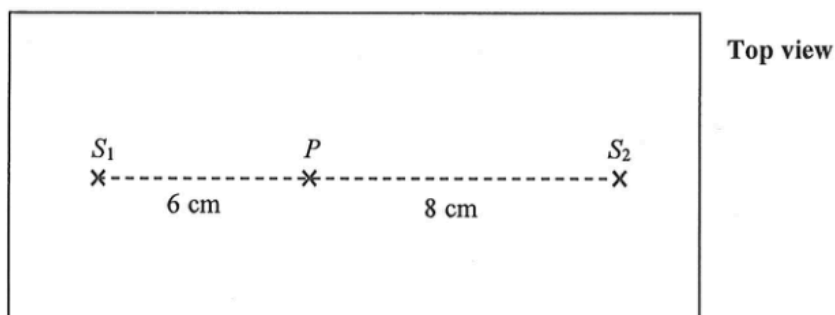
18. A string is tied to a vibrator at one end while the other end is fixed to a wall. The figure shows the stationary wave formed at a certain frequency of the vibrator.



Which statement is correct when the frequency is doubled ?

- A. The amplitude of the wave will double.
- B. The wavelength will double.
- C. The speed of the wave will double.
- D. Particle X and Y will become in phase.

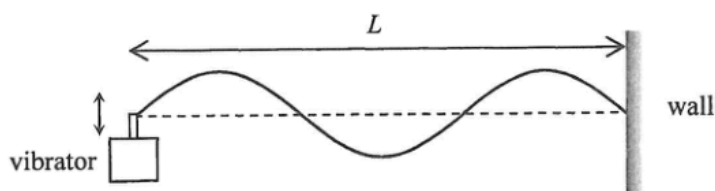
16. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase and generate water waves of wavelength 0.8 cm towards each other at the same speed. For a particle P at distances 6 cm and 8 cm from S_1 and S_2 respectively as shown, the amplitude of the waves reaching P from S_1 is 1 mm while that from S_2 is 2 mm .



What is the amplitude of oscillation of P ?

- A. 0 mm
- B. 1 mm
- C. 2 mm
- D. 3 mm

13. The figure shows a string with one end fixed on the wall and the other end tied to a vibrator with frequency f . A stationary wave is formed as shown at a certain instant. The distance between the vibrator and the wall is L .



Which of the following statements are correct ?

- (1) All particles on the string between two successive nodes are vibrating in phase.
- (2) If the tension in the string increases while L remains unchanged, f must increase to maintain the same stationary wave pattern.
- (3) The wavelength of the stationary wave is $\frac{L}{3}$.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

14. Which of the following phenomena involve(s) the principle of superposition of waves ?

- (1) Ripples in a pond overlap and create regions with different amplitudes.
- (2) Stationary wave patterns are formed when two waves of the same frequency travel in opposite directions.
- (3) The speed of waves changes when they travel from one medium to another.

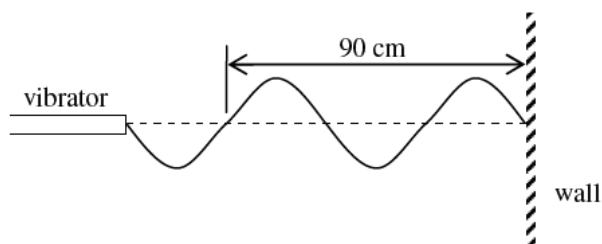
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

19. A monochromatic light beam is directed to a double slit and a fringe pattern is formed. Which of the following physical phenomena is/are involved ?

- (1) dispersion
- (2) diffraction
- (3) interference

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

19.

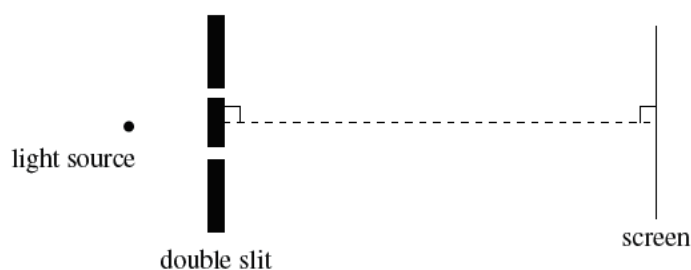


A stationary wave is set up along a string by a vibrator. The waveform at a certain instant is shown above. If the frequency of the vibrator is 50 Hz, what is the wave speed along the string ?

- A. 15 m s^{-1}
- B. 30 m s^{-1}
- C. 45 m s^{-1}
- D. 55 m s^{-1}

pp Q22

*22.

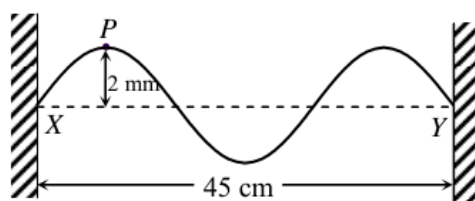


In a Young's double slit experiment, a monochromatic light source of wavelength 600 nm is used. The fringe separation is 5 mm on the screen. If the slit separation is halved and a monochromatic light source of wavelength 450 nm is used instead, what is the new fringe separation ?

- A. 1.9 mm
- B. 3.3 mm
- C. 7.5 mm
- D. 13.3 mm

sap Q19

19.



String XY is fixed at both ends. The distance between X and Y is 45 cm. Two identical sinusoidal waves travel along XY in opposite directions and form a stationary wave with an antinode at point P . The figure shows the string when P is 2 mm, its maximum displacement, from the equilibrium position. What is the amplitude and wavelength of each of the **travelling waves** on the string ?

- | | Amplitude | Wavelength |
|----|-----------|------------|
| A. | 1 mm | 30 cm |
| B. | 1 mm | 15 cm |
| C. | 2 mm | 30 cm |
| D. | 2 mm | 15 cm |

20. A Young's double-slit experiment was performed using a monochromatic light source. Which change would result in a greater fringe separation on the screen ?
- (1) Using monochromatic light source of longer wavelength
 - (2) Using double slit with greater slit separation
 - (3) Using double slit with larger slit width
- A. (1) only
 - B. (1) and (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)